

Track-E

Mobile Activity Tracking Application

Participant	Soham Biswas
Matrikelnummer	321088
Platform	Android 14 / Kotlin
Application package	com.example.track_e

1. What does the app do?

Track-E is a simple Android activity-tracking application. The user can start a tracking session, record walking steps using the phone's accelerometer, and measure travelled distance using GPS/location updates. After a session is stopped, the recorded steps and distance are returned to the home screen and stored locally as a history entry.

The home screen also maintains a daily step total. Each completed session adds its recorded steps to the current day's total, while a new calendar day starts again at zero.

Main functions

- Start and stop an activity-tracking session.
- Detect steps using accelerometer data.
- Calculate travelled distance from GPS/location updates.
- Maintain today's accumulated step count.
- Store and display previous sessions.
- Visualize recorded session step counts with a custom graph.

2. Brief technical notes

The application uses three Activities: MainActivity, TrackingActivity and HistoryActivity. Activity Result API and Intent extras are used to transfer the recorded steps and distance back to the home screen.

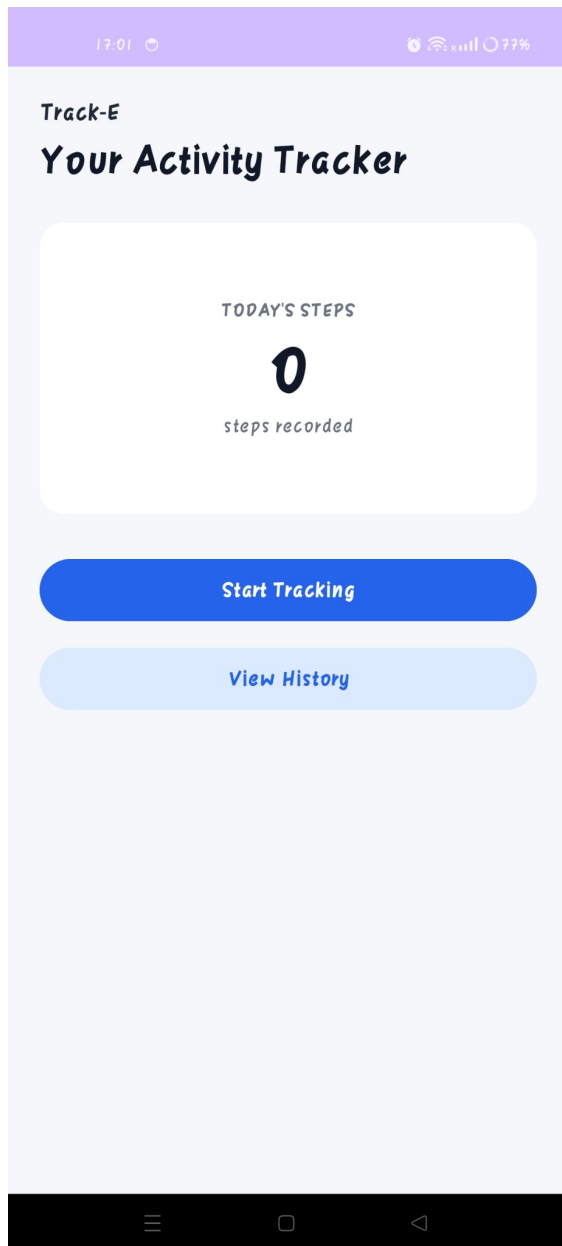
SharedPreferences provides simple local persistence for the history and daily step total.

The step detector is a simple threshold-based accelerometer implementation developed for this project rather than a production-grade pedometer algorithm. GPS distance also depends on the availability and quality of location updates.

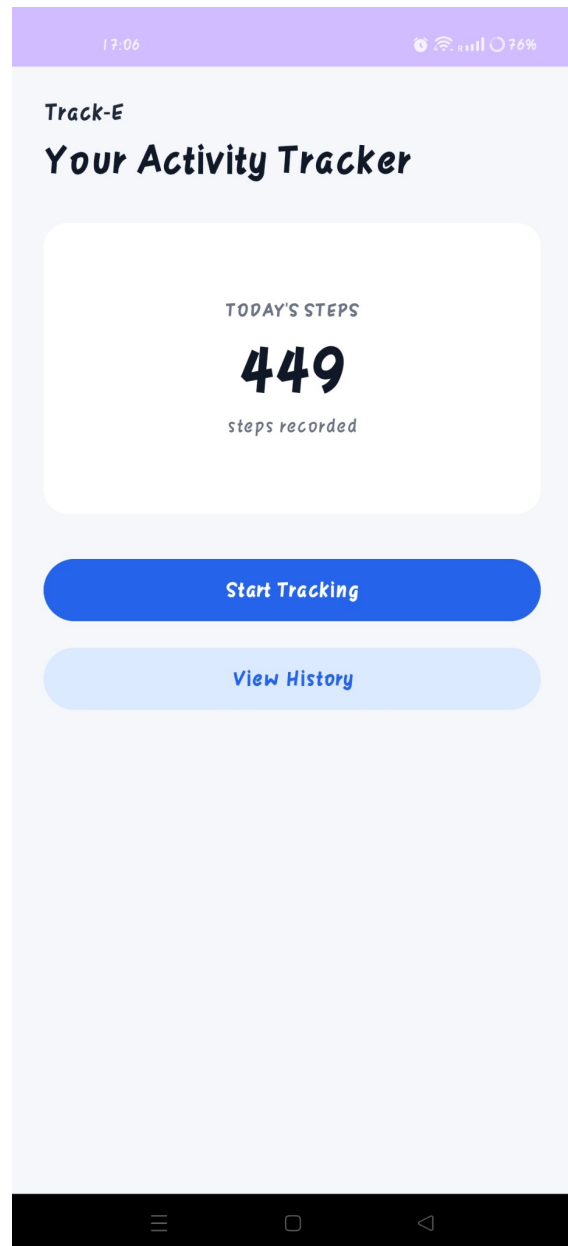
3. How does the app look like?

Home screen and daily step total

The home screen is the main entry point. It shows today's accumulated steps and provides buttons to start a new tracking session or open the activity history.



Initial state: Today's steps starts at 0.

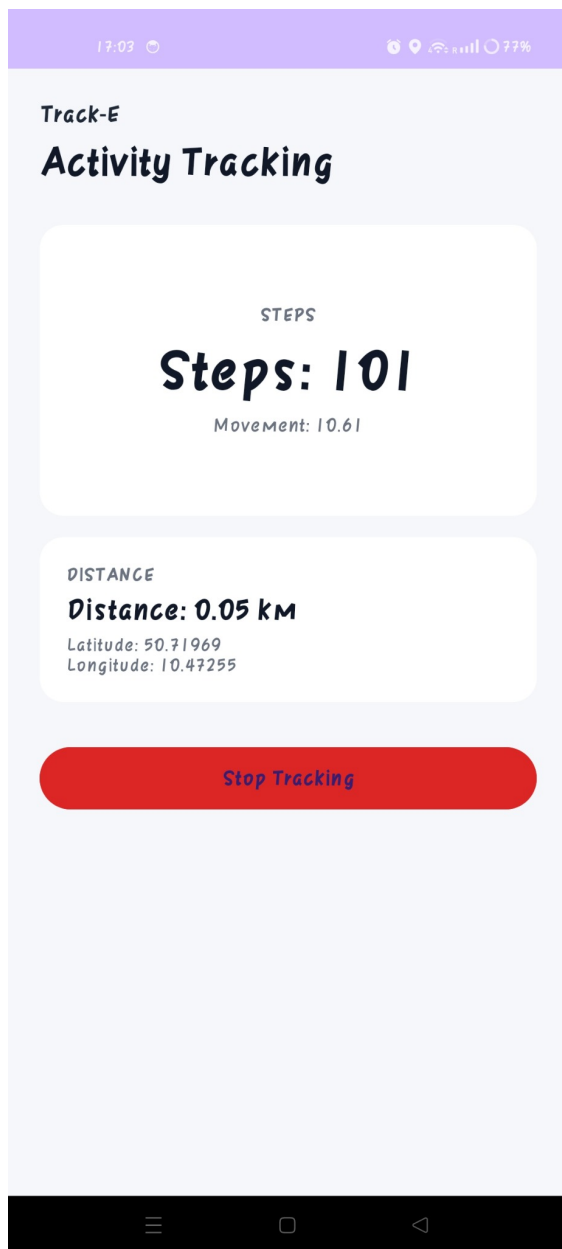


After multiple sessions: Today's steps shows the accumulated total (449 in the example).

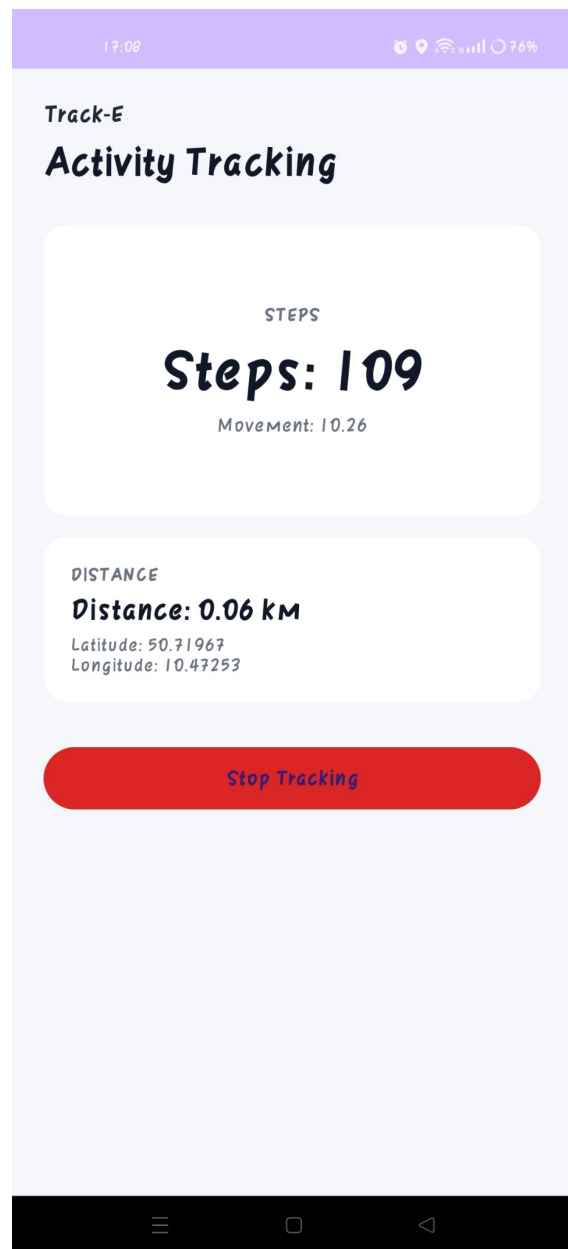
The daily total is independent of the history entries: each tracking session is stored separately, while the home screen adds the session's steps to the current day.

Tracking Activity

During a session, Track-E displays the detected step count, current movement magnitude from the accelerometer, GPS coordinates and accumulated distance. The red button ends the session and returns the results to the home screen.



Example session: 101 steps and 0.05 km.



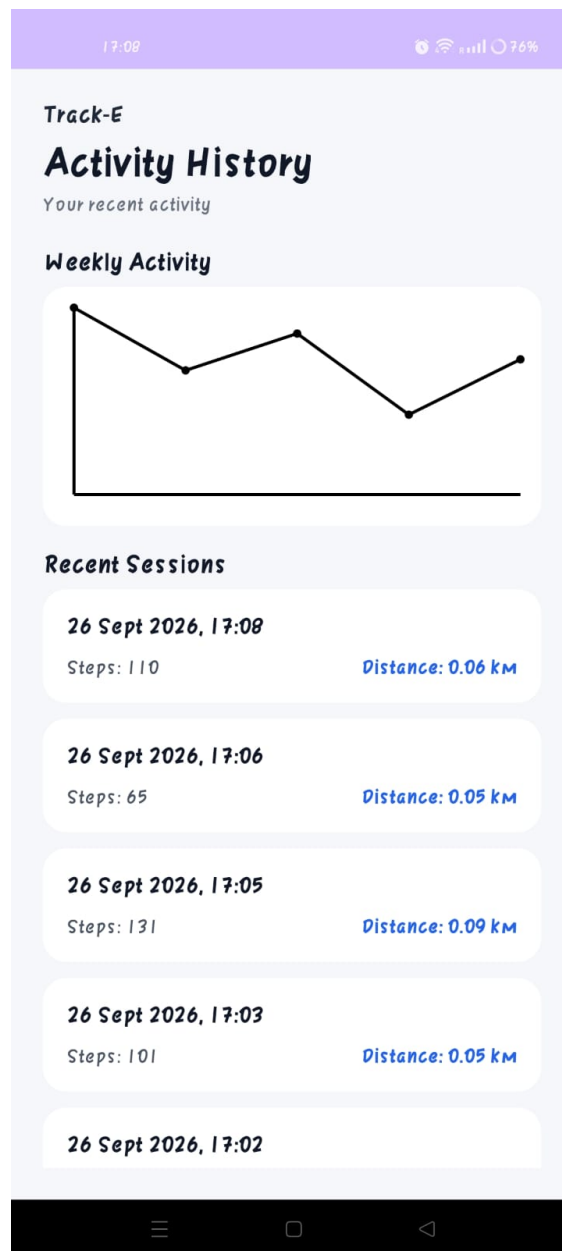
Another session: 109 steps and 0.06 km.

Implementation overview

The accelerometer provides X, Y and Z acceleration values. Track-E calculates their magnitude and compares consecutive readings using a movement threshold and a minimum time interval. GPS updates are received through the fused location provider, and `Location.distanceTo()` is used to accumulate travelled distance.

Activity History

The History screen presents recorded sessions in a RecyclerView and also displays a custom line graph of the recorded step counts. Sessions are shown with their date/time, step count and distance.



Project implementation and assignment elements

Element	Implementation in Track-E
3 Activities	MainActivity, TrackingActivity, HistoryActivity
Activity data transfer	Activity Result API + Intent extras
CustomView	ActivityGraphView draws the activity line graph with Canvas/Paint
RecyclerView	HistoryAdapter displays stored ActivityRecord objects
Two sensor/location inputs	Accelerometer for steps; GPS/location for distance
Local storage	SharedPreferences for history and today's step total
User experience	Separate dashboard, tracking and history screens with clear controls

Conclusion: Track-E demonstrates the required Android concepts in a compact activity-tracking application, combining Activities, Intent-based data exchange, sensor/location input, local persistence, RecyclerView and a custom graphical view.